

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

1. Let m be the Lebesgue measure on $[0, 1]$ and $I_A(x)$ denote the indicator function of A . Consider the following sequence of random variables on $(\Omega = [0, 1], \mathcal{F} = \mathcal{B}[0, 1], P = m)$: For $n \geq 1$, $X_n(\omega) = nI_{(0, \frac{1}{n^2})}(\omega)$ and $X(\omega) = 0$ for $\omega \in \Omega$.

(i) Show $X_n \rightarrow X$ almost surely (P).

(ii) Does $X_n \rightarrow X$ in probability?

(iii) Show $X_n \rightarrow X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ for $0 < p < 2$, but this convergence does not hold for $2 \leq p < \infty$. (7+7+(7+7) = 28 points)

2. Let X, Y be two **independent** random variables on some probability space $(\Omega, \mathcal{F}, P)$, with marginal laws P_X, P_Y on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Recall that X and Y independent iff their joint law is the *product measure* of their marginal laws, i.e $P_{(X,Y)} = (P_X) \times (P_Y)$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let m is the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $m^2 \doteq m \times m$ is the Lebesgue measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let f, g be two non-negative functions in $L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$.

- (a) If X and Y are independent and $E(X^2) < \infty$, $E(Y^2) < \infty$, then carefully prove

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

- (b) Prove: $\sqrt{1 + (E(X))^2} \leq E\sqrt{1 + X^2} \leq 1 + E(|X|)$.

- (c) Suppose $P_X \ll m, P_Y \ll m$, with $\frac{dP_X}{dm} = f$ and $\frac{dP_Y}{dm} = g$. Then prove

$$P_{(X,Y)} \ll m^2, \quad \frac{dP_{(X,Y)}}{dm^2} = f \cdot g \quad \text{a.e } (m^2)$$

- (d) If f is continuous and bounded on $\mathbb{R}$, so is $f * g$.

- (e) If $\lambda \doteq P_X * P_Y$ Prove $\frac{d\lambda}{dm} = f * g$ a.e (m), f and g are as in (b).

- (f) If Z is a random variable with $P_Z = \lambda$ as in (d) above, then how is Z related to X and Y ? What does the result in (d) say about the law of Z ?

(8+8+8+8+8+(5+4) = 49 points)

3. (i) Show that a *Normal*(0, 1) random variable exists:

In other words, using steps of the Caratheodory extension theorem, exhibit a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \rightarrow \mathbb{R}$, such that its c.d.f. is given by Φ where

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du, \quad x \in \mathbb{R}.$$

(ii) Compute the following Lebesgue integral (Clearly stating any result used)

$$\int_{\Omega} X(\omega) dP(\omega).$$

(iii) For each $n \geq 1$, let μ^n be a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Under what conditions can we say that there is a probability measure P on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ such that $P(A \times \mathbb{R}^\infty) = \mu^n(A)$ for any $A \in \mathcal{B}(\mathbb{R}^n)$, for any $n \geq 1$? Describe a sequence of random variables $\{X_n : n \geq 1\}$ defined on $(\Omega = \mathbb{R}^\infty, \mathcal{F} = \mathcal{B}(\mathbb{R}^\infty), P)$ such that the joint distribution of $(X_1, \dots, X_n)$ is given by μ^n , for all $n \geq 1$.

(iv) Using part (iii) or otherwise, show that there exists $\{X_n : n \geq 1\}$ which is an i.i.d. sequence of $Normal(0, 1)$ random variables. (4×7= 28 points)

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^1 p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1, n \in \mathbb{N}$	$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b]) / (b - a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx, a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha, \beta)} \int_A x^{\alpha-1} (1-x)^{\beta-1} I_{(0, 1)}(x) m(dx),$ where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b), a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x - \gamma)^2 / 2\sigma^2}}{x} I_{(0, \infty)}(x) m(dx)$